Why New Coordinates?

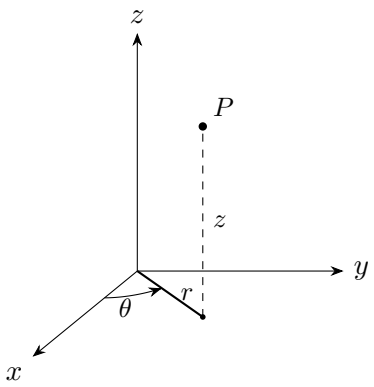
Polar coordinates made double integrals over disks, annuli, and wedges easy: a circle that needs square roots in x and y is just $r = a$. In three dimensions we meet the same problem. In the triple integrals notes we set up the solid between the paraboloid $z = x^2 + y^2$ and the plane $z = 4$, and the bounds $-\sqrt{4 - x^2} \le y \le \sqrt{4 - x^2}$ made the integral painful to evaluate.

There are two natural ways to bring polar ideas into space.

- **Cylindrical coordinates** are polar coordinates in the xy-plane plus the height z. They fit solids that are symmetric about the z-axis: cylinders, paraboloids, cones.
- **Spherical coordinates** measure distance from the origin and two angles. They fit solids that are symmetric about the origin: balls, spherical shells, and regions cut out by cones.

Cylindrical Coordinates

To locate a point $P = (x, y, z)$, drop it straight down to the point $(x, y, 0)$ in the xy-plane. Describe that shadow with polar coordinates (r, θ), and keep the height z. The **cylindrical coordinates** of P are (r, θ, z).



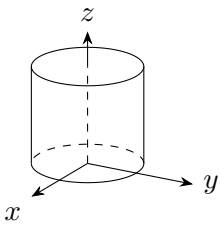
Converting.

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z$$

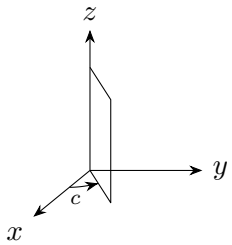
$$r^2 = x^2 + y^2, \qquad \tan\theta = \frac{y}{x}, \qquad z = z$$

As in polar coordinates, we take $r \ge 0$ and $0 \le \theta < 2\pi$, and we use the quadrant of (x, y) to pick the right θ.

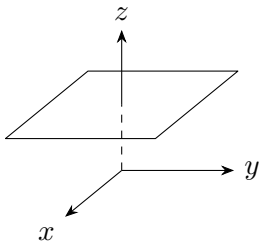
Coordinate surfaces. Setting one cylindrical coordinate equal to a constant gives a simple surface. The first one explains the name.



$r = c$: a cylinder



$\theta = c$: a half-plane



$z = c$: a horizontal plane

Common surfaces in cylindrical coordinates.

	Cylinder	Cone	Sphere	Paraboloid
Rectangular	$x^2 + y^2 = c^2$	$z = c\sqrt{x^2 + y^2}$	$x^2 + y^2 + z^2 = c^2$	$z = c(x^2 + y^2)$
Cylindrical	$r = c$	$z = cr$	$r^2 + z^2 = c^2$	$z = cr^2$

Example 1.

(a) Convert the cylindrical point $(r, \theta, z) = \left(4, \frac{5\pi}{6}, -2\right)$ to rectangular coordinates.

(b) Convert the rectangular point $(x, y, z) = (-1, 1, 3)$ to cylindrical coordinates.

(c) Write the surface $x^2 + y^2 = 2x$ in cylindrical coordinates. What is it?

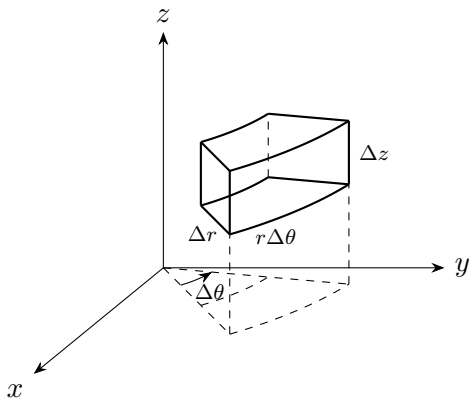
Triple Integrals in Cylindrical Coordinates

Suppose E lies between a bottom surface $z = u_1(x, y)$ and a top surface $z = u_2(x, y)$, over a shadow D in the xy-plane. We already know

$$\iiint_E f\, dV = \iint_D \left[\int_{u_1}^{u_2} f\, dz\right] dA.$$

If D is easy to describe in polar coordinates, we use $dA = r\, dr\, d\theta$ for the outer double integral. Geometrically, a small piece of E is a polar rectangle with sides Δr and $r\,\Delta\theta$, raised to a height Δz, so

$$\Delta V \approx r\,\Delta z\,\Delta r\,\Delta\theta.$$



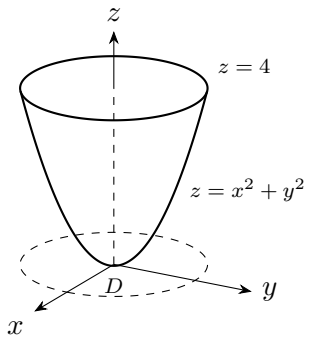
Triple Integrals in Cylindrical Coordinates. If $E = \{(r, \theta, z) \mid \alpha \le \theta \le \beta,\ g_1(\theta) \le r \le g_2(\theta),\ u_1(r, \theta) \le z \le u_2(r, \theta)\}$, then

$$\iiint_E f(x, y, z)\, dV = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} \int_{u_1(r,\theta)}^{u_2(r,\theta)} f(r\cos\theta,\ r\sin\theta,\ z)\, r\, dz\, dr\, d\theta.$$

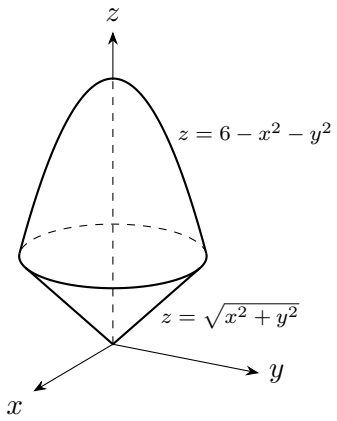
That is, $dV = r\, dz\, dr\, d\theta$. The bounds for z come from the top and bottom surfaces, written in terms of r and θ; the bounds for r and θ describe the shadow D in polar coordinates.

The factor r is the same r as in $dA = r\, dr\, d\theta$. Leaving it out is the most common mistake with cylindrical coordinates.

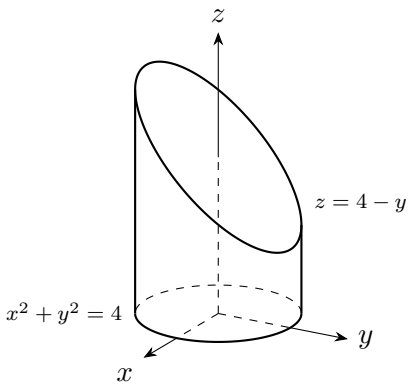
Example 2. Let E be the solid bounded below by the paraboloid $z = x^2 + y^2$ and above by the plane $z = 4$ (the solid we could only set up in the triple integrals notes). Evaluate $\iiint_E \sqrt{x^2 + y^2}\, dV$.



Example 3. Find the volume of the solid E that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the paraboloid $z = 6 - x^2 - y^2$.



Practice 1. Find the volume of the solid inside the cylinder $x^2 + y^2 = 4$, above the plane $z = 0$, and below the plane $z = 4 - y$.



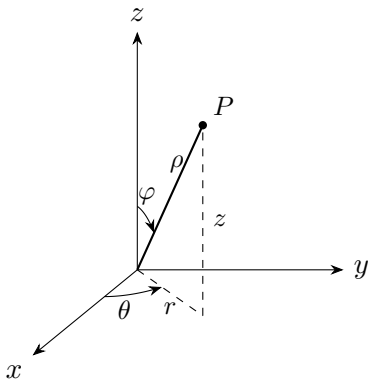
Practice 2. Sketch the shadow D, then convert to cylindrical coordinates and evaluate:

$$\int_{-1}^{1} \int_{0}^{\sqrt{1-x^2}} \int_{0}^{1-x^2-y^2} \left(x^2 + y^2\right) dz\, dy\, dx.$$

Spherical Coordinates

In **spherical coordinates** a point P is described by (ρ, θ, φ), where

- ρ ("rho") is the distance from the origin to P, so $\rho \geq 0$;
- θ is the same angle as in cylindrical coordinates;
- φ ("phi") is the angle from the positive z-axis to the segment OP, so $0 \leq \varphi \leq \pi$.



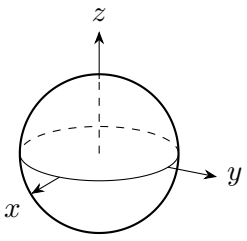
The right triangle with hypotenuse ρ in the figure shows that $z = \rho\cos\varphi$ and $r = \rho\sin\varphi$. Combining $r = \rho\sin\varphi$ with $x = r\cos\theta$ and $y = r\sin\theta$ gives the conversion formulas.

Converting.

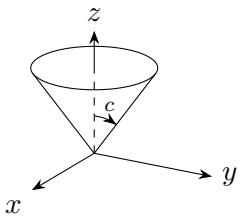
$$x = \rho\sin\varphi\cos\theta, \qquad y = \rho\sin\varphi\sin\theta, \qquad z = \rho\cos\varphi, \qquad \rho^2 = x^2 + y^2 + z^2.$$

Since φ is measured down from the positive z-axis, $\varphi = 0$ points straight up, $\varphi = \pi/2$ points along the xy-plane, and $\varphi = \pi$ points straight down. So φ never needs to go past π. (Some physics books swap the names θ and φ; we follow OpenStax.)

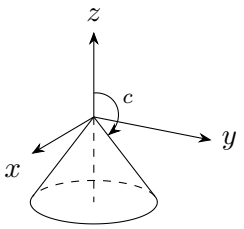
Coordinate surfaces. The surface $\rho = c$ is a sphere of radius c centered at the origin, and $\theta = c$ is the same half-plane as in cylindrical coordinates. The new one is $\varphi = c$: every point makes the same angle with the z-axis, so the surface is a **half-cone** with its vertex at the origin. (The special case $\varphi = \pi/2$ is the xy-plane.)



$\rho = c$: a sphere



$\varphi = c < \frac{\pi}{2}$: cone opening up



$\varphi = c > \frac{\pi}{2}$: cone opening down

Example 4.

(a) Convert the spherical point $(\rho, \theta, \varphi) = \left(4, \frac{\pi}{3}, \frac{\pi}{6}\right)$ to rectangular coordinates.

(b) Write the cone $z = \sqrt{x^2 + y^2}$ in spherical coordinates.

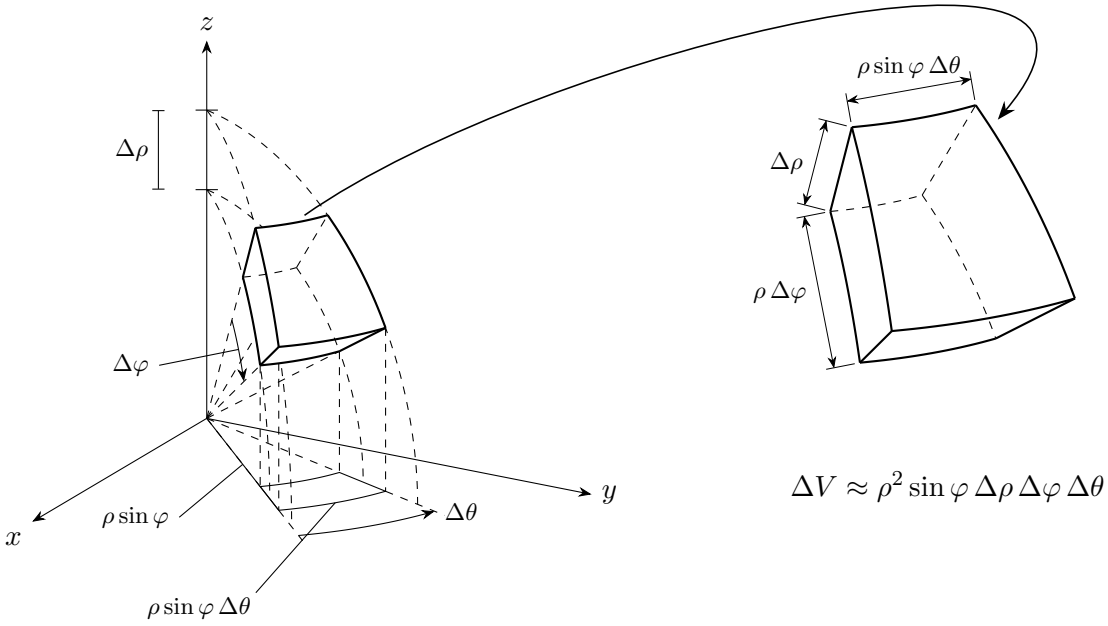
(c) Write the sphere $x^2 + y^2 + z^2 = 2z$ in spherical coordinates.

Triple Integrals in Spherical Coordinates

To find dV, look at a small *spherical box* made by changing ρ, φ, and θ by small amounts. Its three edges are approximately straight:

- the edge in the ρ-direction has length $\Delta\rho$;
- the edge in the φ-direction is an arc of a circle of radius ρ, so its length is $\rho\,\Delta\varphi$;
- the edge in the θ-direction is an arc of a horizontal circle of radius $r = \rho\sin\varphi$, so its length is $\rho\sin\varphi\,\Delta\theta$.

Multiplying the three edge lengths gives $\Delta V \approx \rho^2 \sin\varphi\,\Delta\rho\,\Delta\varphi\,\Delta\theta$.



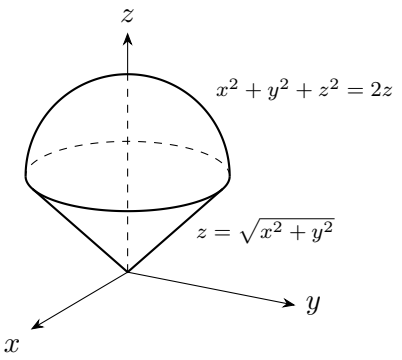
Triple Integrals in Spherical Coordinates. If $E = \{(\rho, \theta, \varphi) \mid \alpha \le \theta \le \beta,\ \gamma \le \varphi \le \psi,\ u_1(\theta, \varphi) \le \rho \le u_2(\theta, \varphi)\}$, then

$$\iiint_E f(x, y, z)\, dV = \int_\alpha^\beta \int_\gamma^\psi \int_{u_1(\theta,\varphi)}^{u_2(\theta,\varphi)} f(\rho \sin\varphi \cos\theta,\ \rho \sin\varphi \sin\theta,\ \rho \cos\varphi)\ \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta.$$

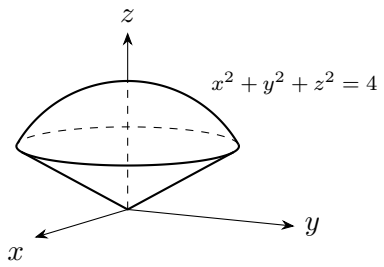
That is, $dV = \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta$. To find the ρ-bounds, picture a ray leaving the origin: it enters E at $\rho = u_1$ and leaves at $\rho = u_2$.

Example 5. Use spherical coordinates to show that a ball of radius a has volume $\frac{4}{3}\pi a^3$.

Example 6. Find the volume of the "ice cream cone" E that lies above the cone $z = \sqrt{x^2 + y^2}$ and inside the sphere $x^2 + y^2 + z^2 = 2z$. (Example 4 converted both surfaces.)

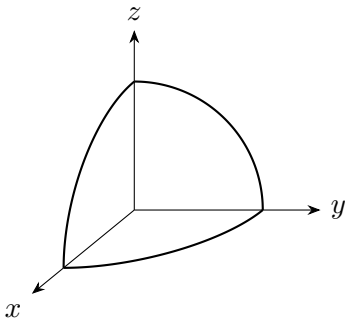


Practice 3. Find the volume of the solid that lies above the cone $z = \sqrt{\frac{x^2+y^2}{3}}$ and inside the sphere $x^2 + y^2 + z^2 = 4$.



Example 7. Convert to spherical coordinates and evaluate:

$$\int_0^2 \int_0^{\sqrt{4-x^2}} \int_0^{\sqrt{4-x^2-y^2}} \left(x^2 + y^2 + z^2\right) dz\, dy\, dx.$$



Choosing a coordinate system.

- **Rectangular:** boxes, and solids bounded by planes or by surfaces like $z = x^2$.
- **Cylindrical:** solids symmetric about the z-axis whose shadow is a disk, annulus, or wedge (cylinders, paraboloids, cones), or integrands containing $x^2 + y^2$.
- **Spherical:** solids bounded by spheres centered at the origin and cones $z = c\sqrt{x^2 + y^2}$, or integrands containing $x^2 + y^2 + z^2$.

Practice 4. For each solid, choose a coordinate system, set up the volume as a triple integral, and evaluate.

(a) The solid inside the sphere $x^2 + y^2 + z^2 = 4$ and outside the cylinder $x^2 + y^2 = 1$.

(b) The solid between the spheres $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2 = 4$, above the xy-plane.

Answers

Practice 1. $\displaystyle\int_0^{2\pi}\int_0^{2}\int_0^{4-r\sin\theta} r\,dz\,dr\,d\theta = 16\pi$. (The $\sin\theta$ term integrates to 0 over $[0, 2\pi]$.)

Practice 2. D is the upper half of the unit disk. $\displaystyle\int_0^{\pi}\int_0^{1}\int_0^{1-r^2} r^2 \cdot r\,dz\,dr\,d\theta = \frac{\pi}{12}$.

Practice 3. The cone is $\varphi = \frac{\pi}{3}$ and the sphere is $\rho = 2$: $\displaystyle\int_0^{2\pi}\int_0^{\pi/3}\int_0^{2} \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = \frac{8\pi}{3}$.

Example 7 (if not finished in class). $\displaystyle\int_0^{\pi/2}\int_0^{\pi/2}\int_0^{2} \rho^2 \cdot \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = \frac{16\pi}{5}$.

Practice 4.

(a) Cylindrical: $\displaystyle\int_0^{2\pi}\int_1^{2}\int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} r\,dz\,dr\,d\theta = 4\sqrt{3}\,\pi$.

(b) Spherical: $\displaystyle\int_0^{2\pi}\int_0^{\pi/2}\int_1^{2} \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = \frac{14\pi}{3}$.